Derivatives of Inverse and Inverse-Trigonometric Functions

Skills practised: Chain rule on arc-functions, the inverse-function derivative, products and quotients, and second derivatives.

Worked pattern. $\dfrac{d}{dx}\arctan(u) = \dfrac{u'}{1+u^2}$, so for $y = \arctan(x^3)$ take $u = x^3$, $u' = 3x^2$, and write $y' = \dfrac{3x^2}{1+x^6}$ — substitute the inner function into the formula, then multiply by its derivative. The same shape works for every arc-function; only the denominator changes.

Leave answers unsimplified only where simplification is genuinely impossible; state exact values (no decimals). Show the inner function u you used.

1. Differentiate: $y = \arctan(3x)$

Answer: ________________

2. Differentiate: $y = \arcsin(x^2)$

Answer: ________________

3. Let $f(x) = x^3 + 2x + 1$. The function f is increasing, so f^{-1} exists.

(a) Find the value b with $f(b) = 4$. **(b)** Use $\left(f^{-1}\right)'(a) = \dfrac{1}{f'(b)}$ to compute $\left(f^{-1}\right)'(4)$.

Answer: ____________________

4. Differentiate: $y = x \arctan(x)$

Answer: ____________________

5. Differentiate: $y = \arccos(2x)$

(Watch the sign that separates arccos from arcsin.)

Answer: ____________________

6. Find the second derivative: $y = \arctan(x)$, find $\dfrac{d^2y}{dx^2}$

Answer: ________________

7. Let $g(x) = 2x + \sin(x)$, which is increasing for all x, so g^{-1} exists. Compute $\left(g^{-1}\right)'(0)$. (Find the input that g sends to 0 first — no algebra needed for this one.)

Answer: ________________

8. Differentiate: $y = e^x \arctan(x)$

Answer: ________________

9. Differentiate: $y = \arctan(\sqrt{x})$
(Simplify the denominator completely.)

Answer: ________________

10. **Challenge.** Find the second derivative of $y = \arctan(2x)$, and simplify it to a single fraction.

Answer: ________________